

# Robot Framework - Installation & Setup

A complete guide to prepare the environment, install, configure, validate and perform the first functional use of the tool.

## Manual objective

This manual guides you through the complete installation and setup process, from preparing the environment to final validation. Follow the steps in order and use the checks throughout the guide to confirm that each configuration was completed correctly.

## Official references

<https://robotframework.org/>

<https://docs.robotframework.org/>

<https://robotframework.org/robotframework/latest/RobotFrameworkUserGuide>

## Recommended virtual environment

Create a project folder, run `python -m venv .venv` and activate it. Use `.venv\Scripts\Activate.ps1` on PowerShell or `source .venv/bin/activate` on macOS/Linux.

## First suite

Create a `.robot` file with Settings, Test Cases and Keywords and run `robot path/to/suite.robot`. Review `output.xml`, `log.html` and `report.html`.

## Troubleshooting

Check PATH, active virtual environment, Python version and whether packages were installed into the correct interpreter.

## How to start safely

Start from the official Robot Framework site and documentation. Confirm prerequisites, supported versions, operating system and required permissions. Learn in a controlled environment before using it in a critical workflow.

## First QA-oriented use

Pick one small, verifiable task in Robot Framework, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

## Common issues and troubleshooting

When Robot Framework fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.

## Installation — prerequisites

Robot Framework is commonly installed with Python. Validate `python --version` and `python -m pip --version` and prefer a virtual environment.

## Install Robot Framework

With the virtual environment active run `python -m pip install robotframework` and validate `robot --version`.

## First test

Create `example.robot` with `*** Test Cases ***` and a simple Log keyword. Run `robot example.robot` and open generated `log.html/report.html`.

## Troubleshooting

If robot is not recognized, confirm the virtual environment is active and Scripts is available. You can also use `python -m robot --version`.